Diabetes Dataset Exploratory Data Analysis & Predictive Modelling

Programming and Database — Data Science

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This project is purely educational. Results must not be used for medical diagnosis or treatment.

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1 Introduction

This report documents a complete data science pipeline applied to a real-world diabetes dataset. The project was developed as part of the *Programming and Database* course of the Data Science Master's programme at the University of Verona (A.Y. 2024/2025).

The work is organised into four main phases:

1. **Data cleaning and preprocessing** — handling null values, encoding categorical features, and detecting physiologically implausible outliers.
2. **Exploratory Data Analysis (EDA)** — understanding feature distributions, class balance, and inter-feature correlations.
3. **Predictive modelling** — training and comparing four classification algorithms on both the original (imbalanced) and an undersampled (balanced) training set.
4. **Interactive web application** — a Streamlit app that exposes the full pipeline and allows real-time predictions from user-supplied patient data.

1.1 Dataset

The raw dataset `diabetes_unclean.csv` was sourced from Kaggle and contains 1 009 patient records with 14 columns. Table 1 describes each feature.

Table 1: Dataset features and descriptions.

Feature	Description
Gender	Patient gender (M/F)
AGE	Age in years
Urea	Blood urea level (mmol/L) — kidney function indicator
Cr	Creatinine level (ţmol/L) — kidney function indicator
HbA1c	Glycated haemoglobin (%) — average blood glucose over 3 months
Chol	Total cholesterol (mmol/L)
TG	Triglycerides (mmol/L)
HDL	High-density lipoprotein (mmol/L) — "good cholesterol"
LDL	Low-density lipoprotein (mmol/L) — "bad cholesterol"
VLDL	Very low-density lipoprotein (mmol/L)
BMI	Body Mass Index (kg/m^2)
CLASS	Target: 1 = diabetic, 0 = non-diabetic

2 Data Cleaning and Preprocessing

2.1 Null Values

The raw dataset contained a small number of missing values (less than 2% of rows). Given this negligible proportion, affected rows were simply dropped rather than imputed,

preserving data integrity without significant information loss.

2.2 Categorical Encoding

Two columns required encoding before numerical analysis could proceed:

- **Gender** — inconsistent casing (`m`, `M`, `f`, `F`) was normalised and mapped to 0 = Male, 1 = Female.
- **CLASS** — contained mixed values (`Y`, `N`, `P`, padded spaces). After stripping and uppercasing, `P` was remapped to `Y`, then encoded as 1 = Diabetic, 0 = Non-diabetic.

2.3 Removal of Irrelevant Columns

Columns `ID` and `No_Pation` are administrative identifiers with no predictive value and were therefore removed.

2.4 Outlier Detection

A standard-deviation criterion was adopted to flag features with potentially anomalous variance:

$$\text{Flag if:} \quad \sigma > \frac{\mu}{2}$$

This identified five features with abnormally high dispersion: **Urea**, **Cr**, **TG**, **HDL**, and **VLDL**.

Physiologically motivated thresholds (Table 2) were then applied to replace values outside biologically plausible ranges with `NaN`, which were subsequently dropped.

Table 2: Physiological thresholds used for outlier removal. *Disclaimer: values obtained from online sources for educational purposes only; not medically validated.*

Feature	**Lower limit**	**Upper limit**	**Unit**
Urea	1	25	mmol/L
Cr	10	400	ţmol/L
TG	0.1	10	mmol/L
HDL	0.3	5	mmol/L
VLDL	0.05	40	mmol/L

Figure 1 shows the distribution of flagged features with threshold lines superimposed.

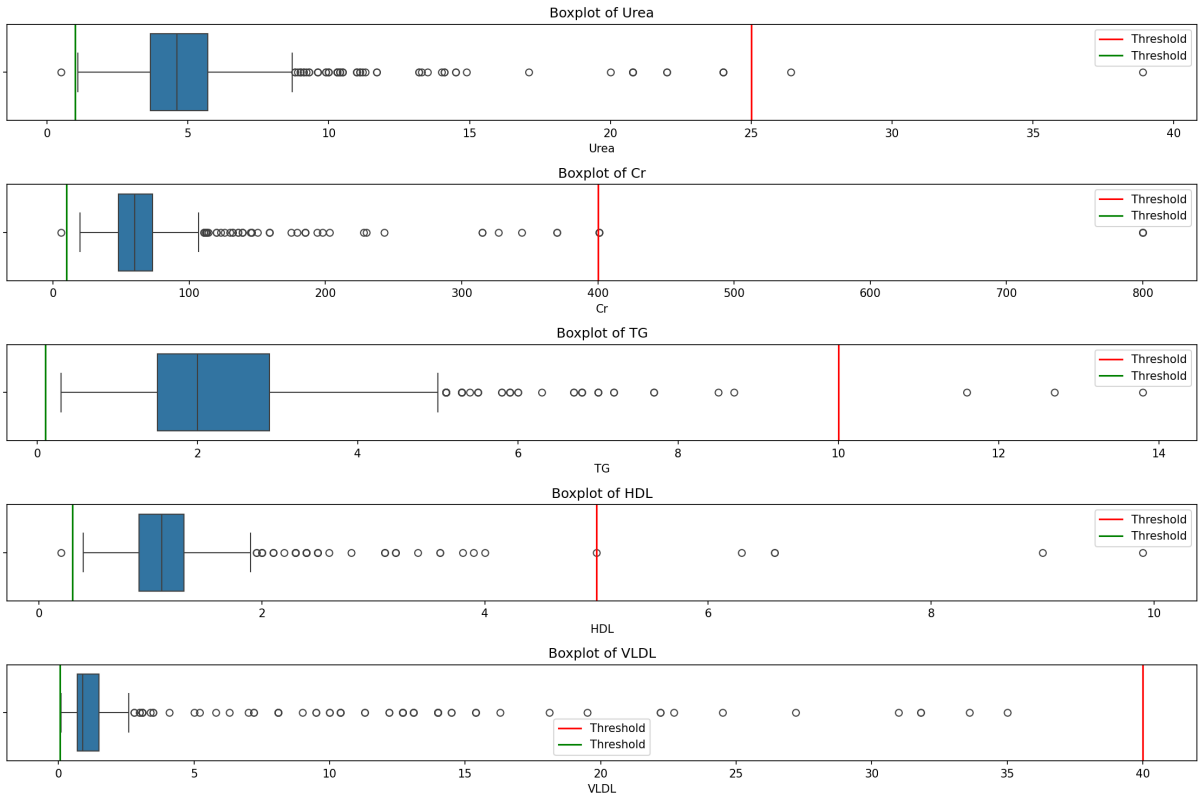
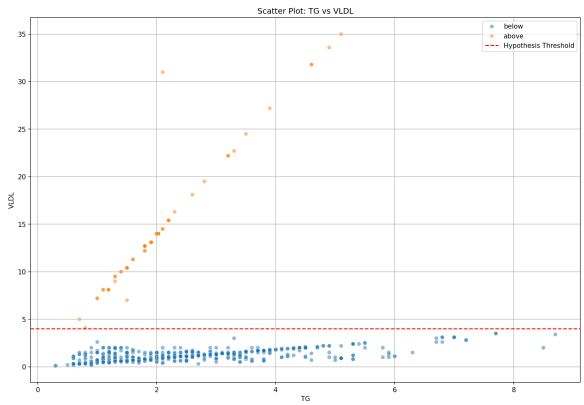


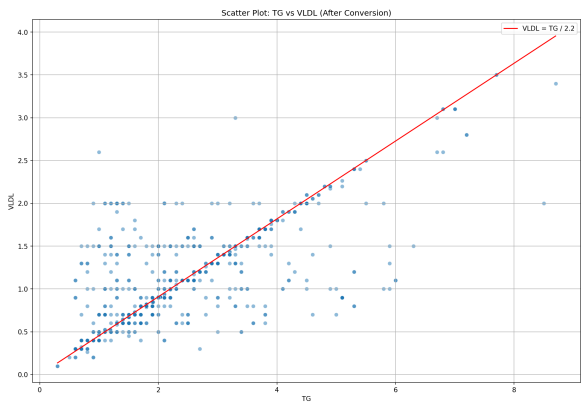
Figure 1: Horizontal boxplots for the five high-variance features. Red lines mark upper physiological limits; green lines mark lower limits.

2.5 VLDL Unit Inconsistency

Scatter plots of TG versus VLDL revealed two distinct clusters (Figure 2a): one group following the expected synthetic relationship VLDL = TG/2.2 (mmol/L), and a second group with values above 4 that followed a different scaling — later identified as measurements recorded in mg/dL rather than mmol/L.



(a) Before correction: two distinct groups visible.



(b) After unit correction: all values follow VLDL = TG/2.2.

Figure 2: TG vs. VLDL scatter plots before and after unit correction.

The conversion applied to values VLDL > 4 was:

$$\text{VLDL}_{\text{mmol/L}} = \frac{\text{VLDL}_{\text{mg/dL}} \times 5.5}{38.67 \times 2.2}$$

After all cleaning steps the dataset comprised **974 patient records** ready for analysis.

3 Exploratory Data Analysis

3.1 Class Distribution

The cleaned dataset is highly imbalanced: 89.5% of patients (872 out of 974) are diabetic (CLASS = 1), while only 10.5% (102 patients) are non-diabetic (CLASS = 0). This imbalance directly motivates the need for dataset balancing before model training, discussed in Section 4.2.

3.2 Pairplot Overview

Figure 3 shows a pairplot of all features coloured by class label, providing a high-level overview of inter-feature relationships and univariate distributions.

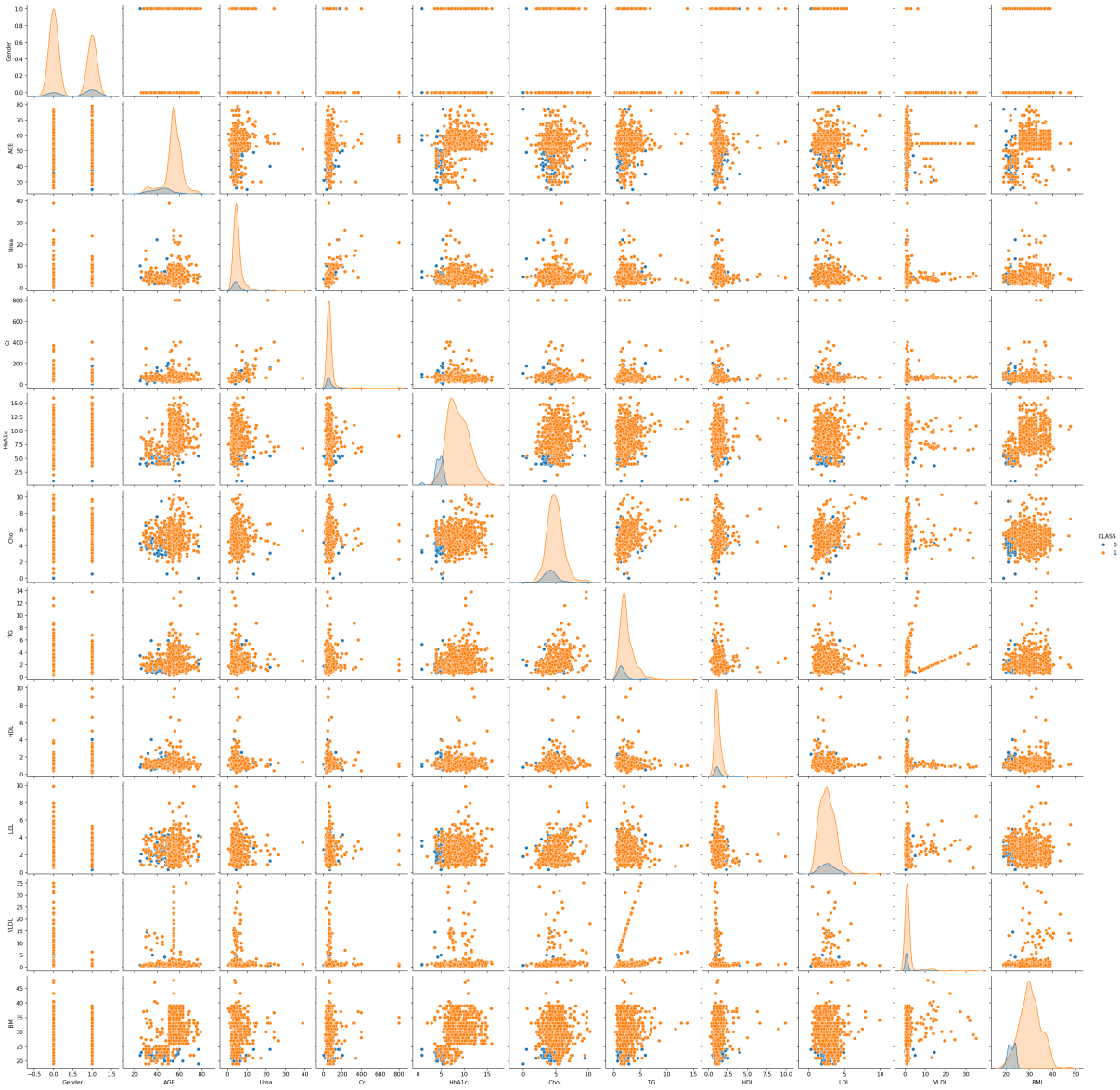


Figure 3: Pairplot of all features, coloured by diabetes status. KDE plots along the diagonal.

3.3 Feature Distributions

Figure 4 presents horizontal boxplots for all numeric features, offering a concise summary of their spread and the presence of residual extreme values after outlier removal.

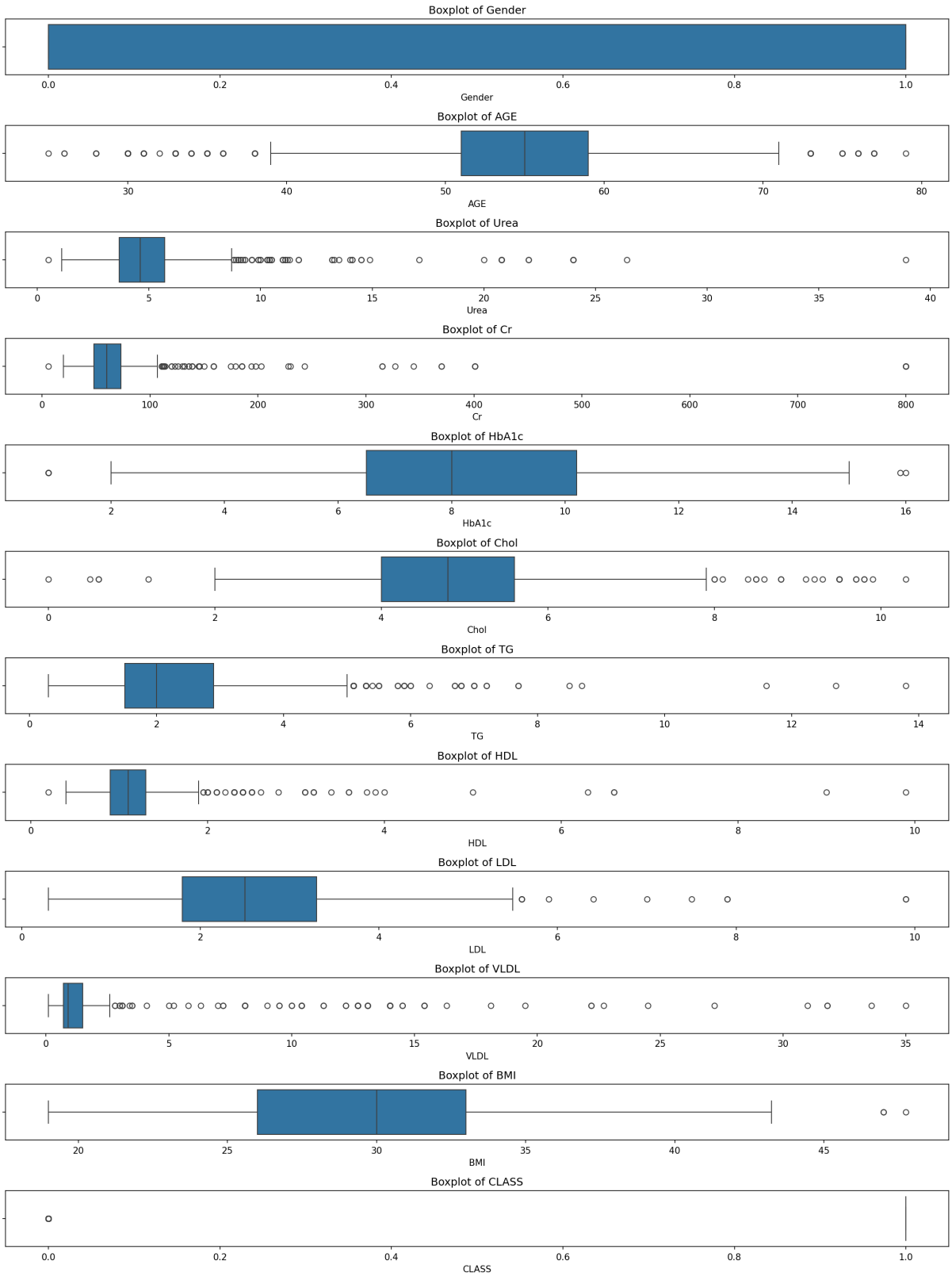


Figure 4: Boxplots of all features in the cleaned dataset.

3.4 Correlation Analysis

The correlation heatmap (Figure 5) highlights the linear associations between features. The strongest correlations with the target variable CLASS are:

- **HbA1c** — the most predictive feature, directly measuring average blood glucose levels.
- **BMI** — a strong secondary predictor; higher BMI is a known risk factor for type 2 diabetes.
- **AGE** — older patients show a higher prevalence of diabetes in this dataset.

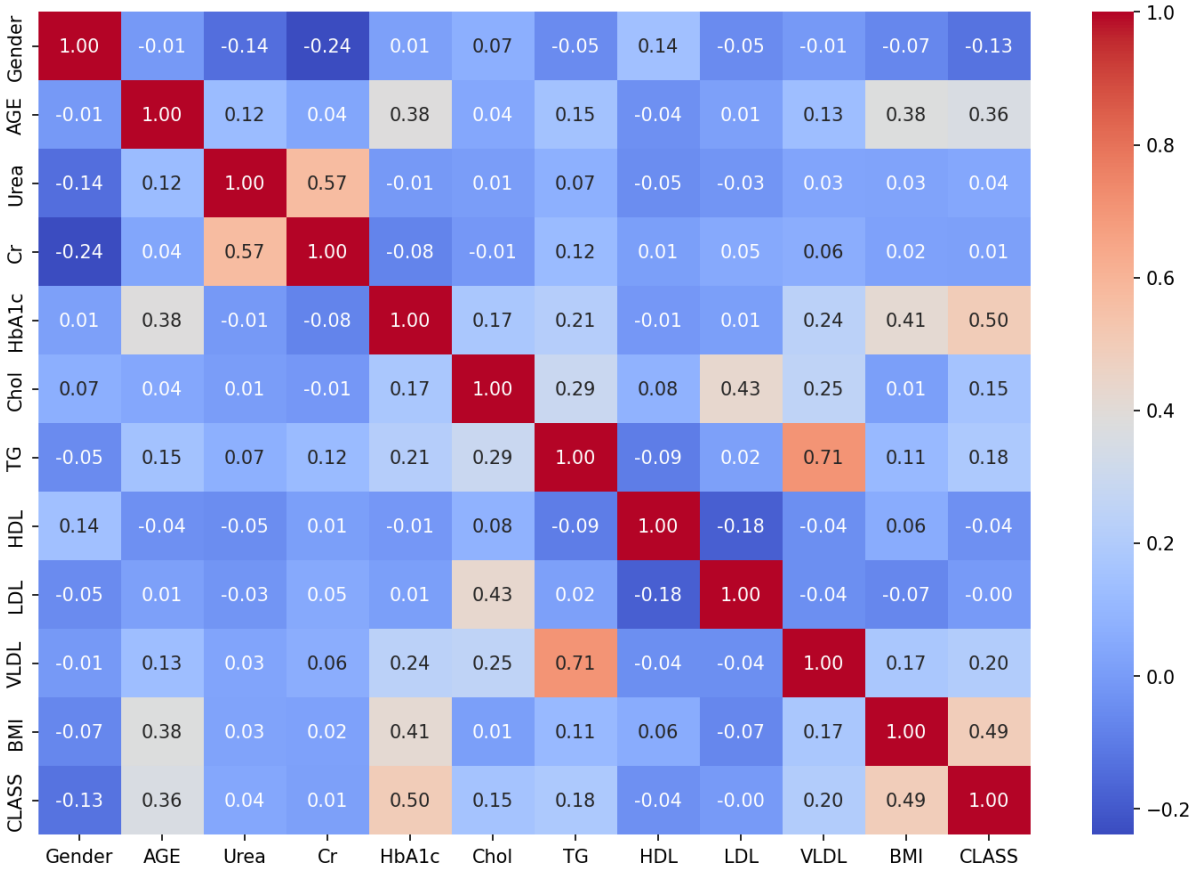


Figure 5: Pearson correlation heatmap of all numeric features.

4 Predictive Modelling

4.1 Train/Test Split

The dataset was split into a training set (80%, $n = 779$) and a held-out test set (20%, $n = 195$) using stratified sampling to preserve the class ratio.

4.2 Class Balancing via Undersampling

Training directly on the imbalanced set risks producing models biased towards predicting the majority class (diabetic). To mitigate this, **random undersampling** of the majority class was applied to the training set: 82 samples were drawn at random from the 697 diabetic training records to match the 82 non-diabetic records, yielding a balanced training set of 164 samples with a 50/50 split.

4.3 Models

Four classification algorithms were evaluated:

- **Logistic Regression** — linear probabilistic classifier.
- **Decision Tree** — non-linear, tree-based, highly interpretable.
- **K-Nearest Neighbours (KNN)** — instance-based, captures local patterns.
- **Random Forest** — ensemble of decision trees; robust and typically high-performing.

Each model was trained and evaluated under both the imbalanced and balanced training conditions, always tested on the same held-out test set.

4.4 Results

Tables 3 and 4 report the performance metrics on the test set. Figure 6 shows the metrics comparison for the balanced training set visually, while Figure 7 tracks how each metric changes from imbalanced to balanced training.

Table 3: Test-set performance metrics — **balanced** training set.

Model	Accuracy	Precision	F1-Score	Recall (0)	Recall (1)
Logistic Regression	0.959	0.994	0.977	0.950	0.960
Decision Tree	**0.995**	**1.000**	**0.997**	**1.000**	0.994
KNN	0.872	0.975	0.925	0.800	0.880
Random Forest	0.985	**1.000**	0.991	**1.000**	0.983

Table 4: Test-set performance metrics — **imbalanced** training set.

Model	Accuracy	Precision	F1-Score	Recall (0)	Recall (1)
Logistic Regression	0.959	0.972	0.977	0.750	0.983
Decision Tree	**0.995**	0.994	**0.997**	0.950	**1.000**
KNN	0.923	0.935	0.958	0.400	0.983
Random Forest	**0.995**	0.994	**0.997**	0.950	**1.000**

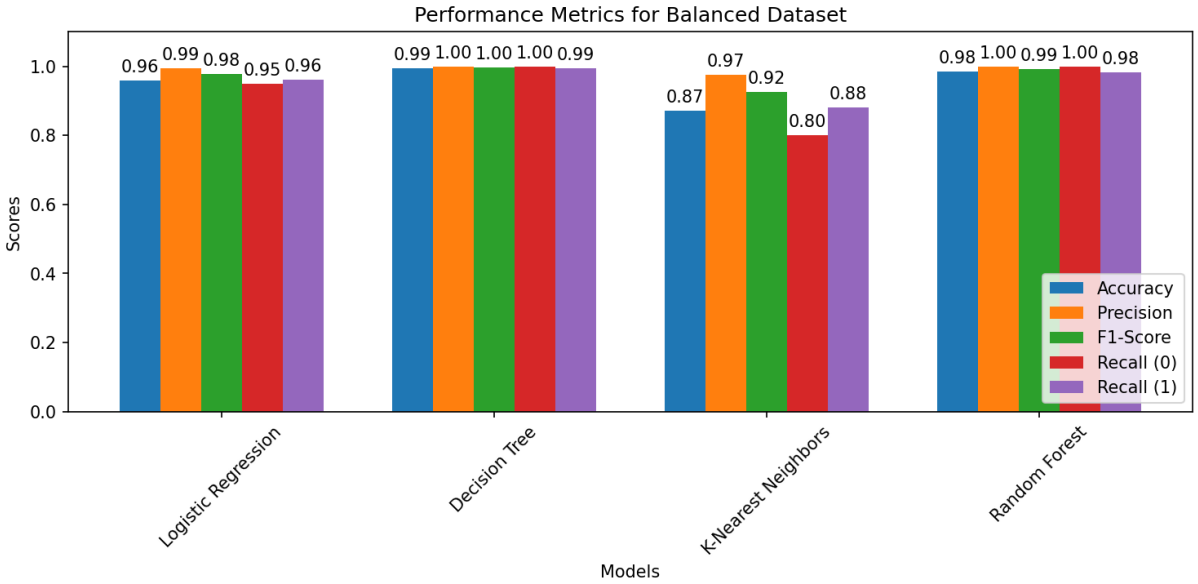


Figure 6: Grouped bar chart of all metrics for each model trained on the balanced dataset.

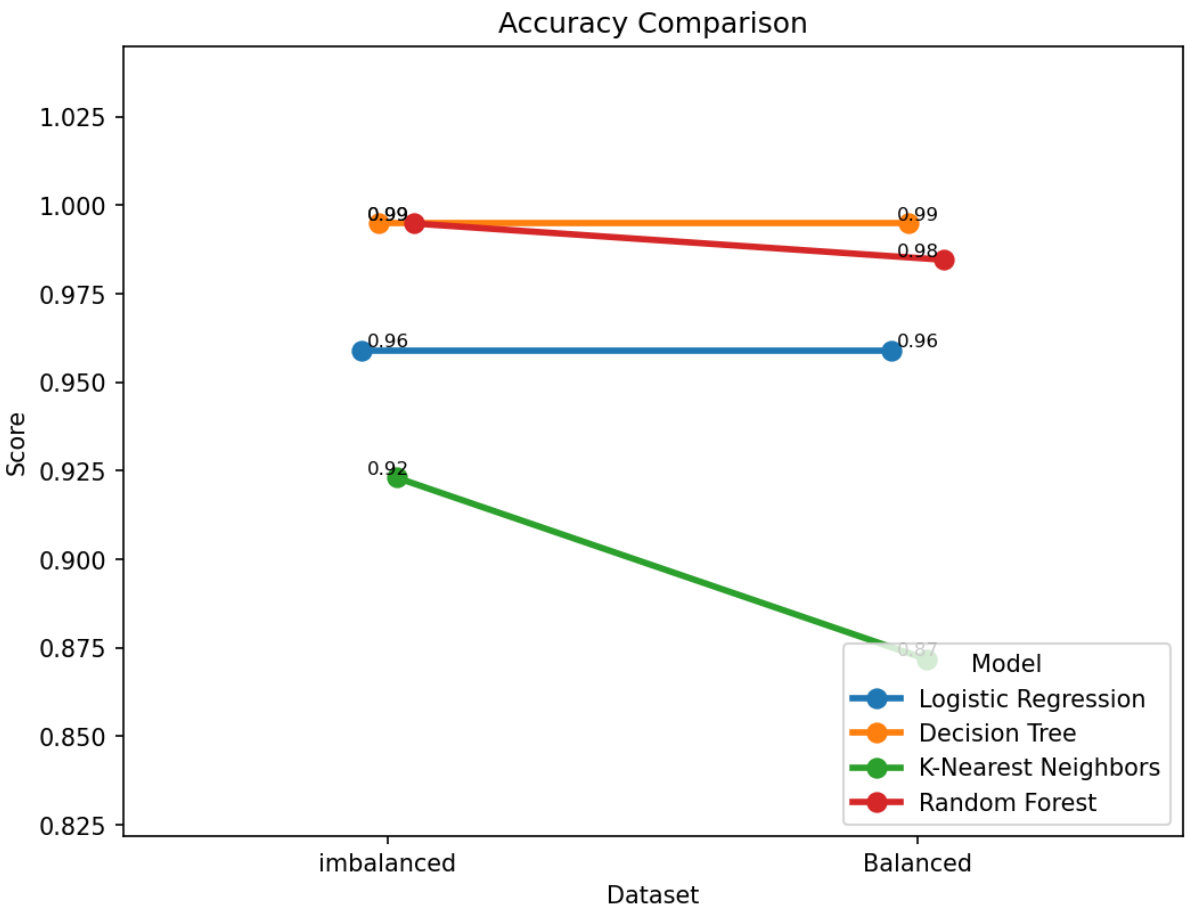


Figure 7: Accuracy comparison across models: imbalanced vs. balanced training.

4.5 Confusion Matrices

Figures 8 and 9 display the confusion matrices for all four models under both training conditions.

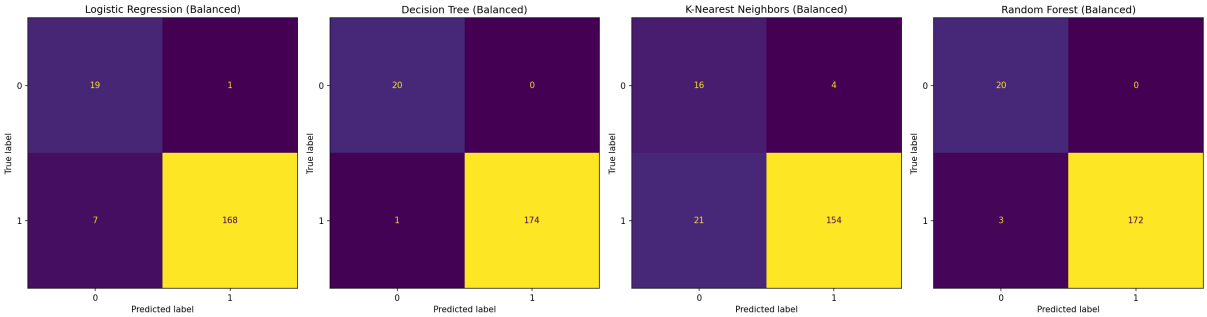


Figure 8: Confusion matrices — models trained on the **balanced** training set.

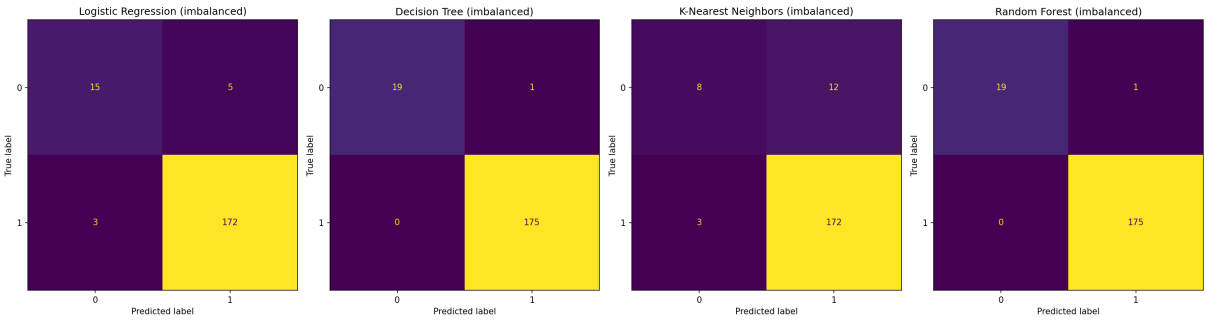


Figure 9: Confusion matrices — models trained on the **imbalanced** training set.

4.6 Feature Importance

Figure 10 shows the feature importances extracted from the Random Forest model trained on the balanced dataset. **HbA1c** is by far the most discriminative feature, confirming the EDA findings. BMI and AGE rank second and third, consistent with the correlation analysis.

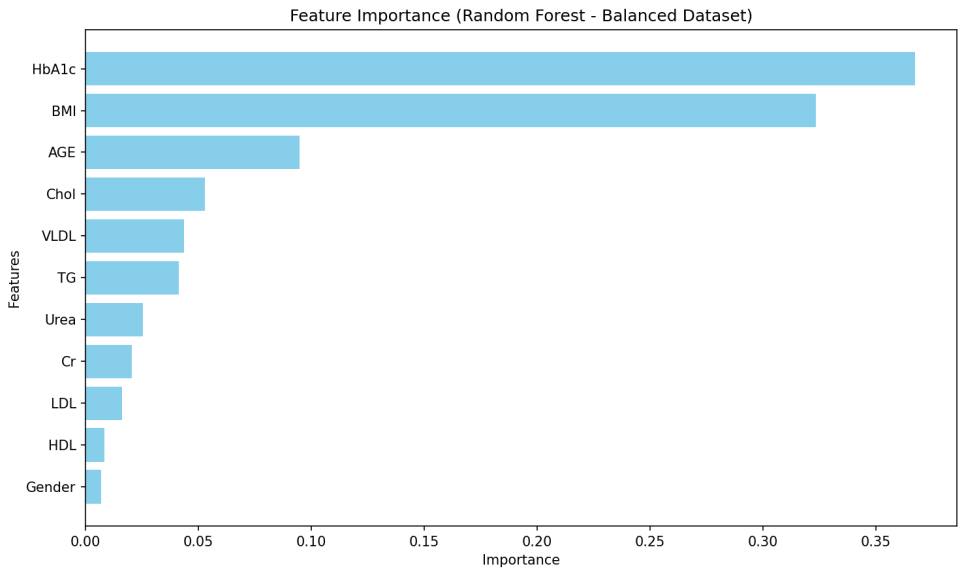


Figure 10: Feature importances from the Random Forest classifier (balanced training set).

4.7 Discussion

The key observations from the modelling phase are:

- **Decision Tree** and **Random Forest** achieve near-perfect performance (accuracy = 99.5%, $F_1 \approx 0.997$) on this dataset under both training conditions. The high scores are partly attributable to the strong separability provided by HbA1c.
- **Balancing improves minority-class recall**: the most visible impact of under-sampling is on *Recall (0)* — i.e., the ability to correctly identify non-diabetic patients. Logistic Regression's Recall (0) improves from 0.750 (imbalanced) to 0.950 (balanced), and KNN's from 0.400 to 0.800. This is clinically significant: failing to identify a true non-diabetic is a false positive that carries real consequences.
- **KNN** is the weakest performer across all conditions, likely because it is sensitive to the feature scale and the high class imbalance in the original data.

5 Streamlit Web Application

An interactive web application built with Streamlit was developed to make the full pipeline accessible without requiring any programming knowledge. The app is located in `src/ST_EDA_prediction.py` and can be launched with:

```
cd src
streamlit run ST_EDA_prediction.py
```

The application is divided into the following sections:

- **Dataset Overview** — raw data preview and summary statistics.
- **Data Cleaning** — step-by-step walkthrough of the cleaning pipeline, with live tables and charts for each step.
- **Outlier Detection** — interactive boxplots with physiological threshold lines.

- **VLDL/TG Analysis** — scatter plots illustrating the unit-inconsistency discovery and correction.
- **EDA** — gender distribution, class distribution, age KDE, interactive feature selector, scatterplot builder, correlation heatmap, and key-feature boxplots by class.
- **Model Training & Benchmarking** — trains all four models, displays metric tables and pointplots for both balanced and imbalanced conditions.
- **Prediction** — a form for entering patient data (age, gender, blood markers, BMI) and receiving real-time predictions from all four models simultaneously.

6 Conclusions

This project demonstrated a complete end-to-end data science workflow on a real healthcare dataset:

1. A thorough cleaning pipeline resolved missing values, encoding inconsistencies, physiologically implausible outliers, and a unit-measurement error in the VLDL column.
2. EDA confirmed that **HbA1c**, **BMI**, and **AGE** are the most informative predictors of diabetes in this dataset.
3. Among the four classifiers evaluated, **Decision Tree** and **Random Forest** achieved the best overall performance ($F_1 \approx 0.997$). Balancing the training set is crucial for improving recall on the minority (non-diabetic) class - particularly relevant in a medical screening context.
4. The Streamlit application provides an intuitive interface for non-technical users to explore the data and obtain real-time predictions.

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